



剑桥雅思11 | 听力真题 Test 4 Section 4

【剑桥雅思11 | 听力真题 Test 4 Section 4 | 土壤碳革命：农业如何成为气候救星？】

Summary: Scientists propose enhancing soil's natural ability to absorb CO2 through regenerative agriculture, potentially offsetting 13% of atmospheric carbon. Research in California and Australia shows composting and year-round grass cultivation can significantly boost soil carbon storage. Experts argue farmers should be incentivized for carbon sequestration to combat climate change.

摘要： 科学家提出通过再生农业增强土壤吸收二氧化碳的自然能力，或可抵消大气中13%的碳含量。加州和澳大利亚的研究表明，堆肥和全年种植草类能显著增加土壤碳储量。专家建议通过经济激励推动农业碳封存，以应对气候变化。



IELTS

听9遍 考9分



🕒 Estimated Reading Time: 8 min.

📖 四级生词 📖 六级生词 📖 雅思生词 📖 托福生词 📖 专八生词 📖 SAT生词 📖 考研生词 📖
GRE生词 📖 高考生词 📖 其它生词

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▶ Section 4.

第四部分。

▶ You will hear part of a lecture about a way of reducing the amount of carbon dioxide in the atmosphere.

你将听到一场关于减少大气中二氧化碳含量的讲座的一部分。

▶ First, you have some time to look at questions 31 to 40.

首先，你有时间看一下第31至40题。

▶ Now, listen carefully and answer questions 31 to 40.

现在，请仔细听并回答第31至40题。

▶ As we saw in the last lecture, a major cause of climate change is the rapid rise in the level of carbon dioxide in the atmosphere over the last century.

正如我们在上一场讲座中所看到的，气候变化的一个主要原因是过去一个世纪大气中二氧化碳水平的迅速上升。

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▶ If we could reduce the amount of CO₂, perhaps the rate of climate change could also be slowed down.

如果我们能减少二氧化碳的量，也许气候变化的速度也会减缓。

▶ One potential method involves enhancing the role of the soil that plants grow in with regard to absorbing CO₂.

一种潜在的方法涉及增强植物生长土壤在吸收二氧化碳方面的作用。

▶ Rat and Laugh, a soil scientist from Ohio State University in the USA, claims that the world's agricultural soils could potentially absorb 13% of the carbon dioxide in the atmosphere, the equivalent of the amount released in the last 30 years, and research is going on into how this might be achieved.

来自美国俄亥俄州立大学的土壤科学家Rat和Laugh声称，全球农业土壤可能吸收大气中13%的二氧化碳，相当于过去30年释放的量，目前正在研究如何实现这一点。

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▶ While first came to the idea that soil might be valuable in this way, not through an interesting climate change, but rather out of concern for the land itself and the people dependent on it.

最初想到土壤在这方面可能有价值，并非因为对气候变化的兴趣，而是出于对土地本身和依赖它的人们的关注。

⏮ Carbon rich soil is dark, crumbly and fertile and retains some water, but erosion can occur if soil is dry, which is a likely effect if it contains inadequate amounts of carbon.

富含碳的土壤颜色深、松散且肥沃，并能保留一些水分，但如果土壤干燥，就可能发生侵蚀，这在碳含量不足时很可能发生。

⏮ Carbon is of course bad for people trying to grow crops or breed animals on that terrain.

当然，碳含量不足对试图在该地形上种植作物或饲养动物的人们不利。

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⏮ In the 1970s and 80s Laugh was studying soils in Africa so devoid of organic matter that the ground had become extremely hard, like cement.

在20世纪70年代和80年代，Laugh研究了非洲的土壤，这些土壤缺乏有机质，以至于地面变得非常坚硬，像水泥一样。

⏮ There he met a pioneer in the study of global warming who suggested that carbon from the soil had moved into the atmosphere.

在那里，他遇到了一位研究全球变暖的先驱，后者提出土壤中的碳已经转移到了大气中。

⏮ This is now looking increasingly likely.

现在看来这越来越有可能。

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⏮ Let me explain.

让我解释一下。

⏮ For millions of years, carbon dioxide levels in the atmosphere have been regulated in part by a natural partnership between plants and microbes, tiny organisms in the soil.

数百万年来，大气中的二氧化碳水平部分由植物和土壤中的微生物（微小生物）之间的自然合作关系调节。

⏮ Plants absorb CO₂ from the air and transform it into sugars and other carbon-based substances.

植物从空气中吸收二氧化碳并将其转化为糖和其他碳基物质。

🎧 Similar proportion of these carbon products remain in the plant, some transfer from the roots to fungi and soil microbes which store the carbon in the soil.

这些碳产物中有相似比例留在植物中，部分从根部转移到真菌和土壤微生物中，后者将碳储存在土壤中。

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🎧 The invention of agriculture some 10,000 years ago disrupted these ancient soil-building processes and led to the loss of carbon from the soil.

大约一万年前的农业发明破坏了这些古老的土壤形成过程，并导致土壤中碳的流失。

🎧 When humans started draining the natural top soil and plowing it up for planting, they exposed the buried carbon to oxygen.

当人类开始排干天然表土并翻耕种植时，他们将埋藏的碳暴露于氧气中。

🎧 This created carbon dioxide and released it into the air and in some places grazing by domesticated animals has removed all vegetation releasing carbon into the air.

这产生了二氧化碳并将其释放到空气中，而在某些地方，家畜的放牧已经移除了所有植被，将碳释放到空气中。

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🎧 Tons of carbon have been stripped from the world's soils where it's needed and pumped into the atmosphere.

大量的碳从世界土壤中被剥离，而这些碳本应留在土壤中，却被排放到大气中。

🎧 So, what can be done?

那么，我们能做些什么？

🎧 Organizers are now coming up with evidence that even modest changes to farming can significantly help to reduce the amount of carbon in the atmosphere.

组织者现在提出证据表明，即使对农业进行适度的改变也能显著帮助减少大气中的碳含量。

🎧 Some growers have already started using an approach known as regenerative agriculture.

一些种植者已经开始使用一种称为再生农业的方法。

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▶ This aims to boost the fertility of soil and keep it moist through established practices.

其目的是通过既定做法提高土壤肥力并保持其湿润。

▶ These include keeping fields planted all year round and increasing the variety of plants being grown.

这些做法包括全年保持田地种植并增加种植植物的多样性。

▶ Strategies like these can significantly increase the amount of carbon stored in the soil, so agricultural researchers are now building a case for their use in combating climate change.

这样的策略可以显著增加土壤中储存的碳量，因此农业研究人员现在正在为它们在应对气候变化中的应用建立案例。

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▶ One American investigation into the potential for storing CO₂ on agricultural lands is taking place in California.

一项关于农业用地储存二氧化碳潜力的美国调查正在加利福尼亚进行。

▶ Local scientist Wendy Silver of the University of California Berkeley is conducting a first of its kind study on a large cattle farm in the state.

加州大学伯克利分校的当地科学家Wendy Silver正在该州的一个大型养牛场进行一项首创的研究。

▶ She and her students are testing the effects on carbon storage of the compost that is created from waste, both agricultural, including manure and cornstorks and waste produced in gardens such as leaves, branches and lawn trimmings.

她和她的学生正在测试由废物制成的堆肥对碳储存的影响，这些废物包括农业废物（如粪便和玉米秸秆）以及花园中产生的废物（如树叶、树枝和草坪修剪物）。

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▶ In Australia, soil ecologist Christine Jones is testing another promising soil enrichment strategy.

在澳大利亚，土壤生态学家Christine Jones正在测试另一种有前景的土壤富集策略。

🎧 Jones and 12 farmers are working to build up soil carbon by cultivating grasses that stay green all year round.

Jones和12位农民正在通过种植全年保持绿色的草来增加土壤碳含量。

🎧 Like composting, the approach has already been proved experimentally.

与堆肥一样，这种方法已经在实验中得到了验证。

🎧 Jones now hopes to show that it can be applied on working farms and that the resulting carbon capture can be accurately measured.

Jones现在希望证明它可以应用于实际农场，并且由此产生的碳捕获可以准确测量。

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🎧 It's hoped in the future that projects such as these will demonstrate the role that farmers and other land managers can play in reducing the harmful effects of greenhouse gases.

希望未来这些项目将展示农民和其他土地管理者在减少温室气体有害影响方面可以发挥的作用。

🎧 For example, in countries like the United States where most farming operations use large applications of fertilizer, changing such long standing habits will require a change of system.

例如，在美国等大多数农业操作大量使用化肥的国家，改变这种长期习惯将需要系统的改变。

🎧 Ratan Lau argues that farmers should receive payment, not just for the corn or beef they produce, but also for the carbon they can store in their soil.

Ratan Lau认为，农民应该获得报酬，不仅因为他们生产的玉米或牛肉，还因为他们可以在土壤中储存的碳。

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🎧 Another study being carried out.

另一项研究正在进行中。

🎧 That is the end of section 4.

第四部分到此结束。

▶ You now have half a minute to check your answers.

你现在有半分钟时间检查你的答案。

▶ That is the end of the listening test.

听力测试到此结束。

▶ You now have 10 minutes to transfer your answers to the answer sheet.

你现在有10分钟时间将答案誊写到答题卡上。

▶ Please stop writing and wait for your question booklet to be collected.

请停止书写，等待你的问题册被收走。

